

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Completed all capstone methods: data collection, wrangling, EDA, SQL, maps, dashboard-style analytics, and modeling.
- 90 Falcon 9 launches; observed landing success rate is 66.7%.
- Success improved over time, reaching roughly 84% in 2020.
- Decision Tree had the strongest cross-validation score; all models reached 83.3% test accuracy.

Introduction

- SpaceX can reduce launch cost when Falcon 9 first stages land and are reusable.
- The project predicts first-stage landing success from launch and mission features.
- The analysis combines API data, web scraping, EDA, SQL, maps, and classification models.



Section 1

Methodology

Methodology

- Collected records from SpaceX API and a static Wikipedia launch-table snapshot.
- Wrangled launch, payload, booster, orbit, site, and landing outcome fields.
- Explored results with pandas, seaborn, SQL, Folium, and dashboard-style plots.
- Tuned four classifiers with 10-fold GridSearchCV.

Data Collection

- API data supplied launch metadata, payload data, rocket versions, launchpad coordinates, and core reuse details.
- Web scraping supplied launch-history tables from the June 2021 Wikipedia snapshot.
- Both sources were normalized into pandas DataFrames.

Data Collection - SpaceX API

- Requested launch records from SpaceX v4 endpoint or course static JSON.
- Normalized nested JSON with `pandas.json_normalize`.
- Enriched rocket, launchpad, payload, and core identifiers.
- Notebook: Module 1 API collection notebook.

Data Collection - Scraping

- Requested the static Wikipedia revision with requests.
- Parsed launch tables with BeautifulSoup.
- Extracted date, booster, site, payload, orbit, customer, outcome, and landing status.

Data Wrangling

- Filtered to Falcon 9 records and re-indexed FlightNumber.
- Filled missing PayloadMass with the column mean.
- Created binary Class labels for successful and unsuccessful landing outcomes.

EDA with Data Visualization

- Plotted outcomes by flight number, launch site, payload mass, orbit, and year.
- Used scatter plots for launch-level detail and bar/line plots for aggregate success rates.
- Prepared one-hot encoded categorical features.

EDA with SQL

- Loaded SpaceX CSV into SQLite.
- Queried launch sites, CCA records, payload metrics, booster filters, mission outcomes, date records, and outcome rankings.

Build an Interactive Map with Folium

- Added launch-site circles and labels.
- Added clustered launch markers colored by outcome.
- Calculated distance lines from CCAFS LC-40 to coastline, railway, highway, and city points.

Build a Dashboard with Plotly Dash

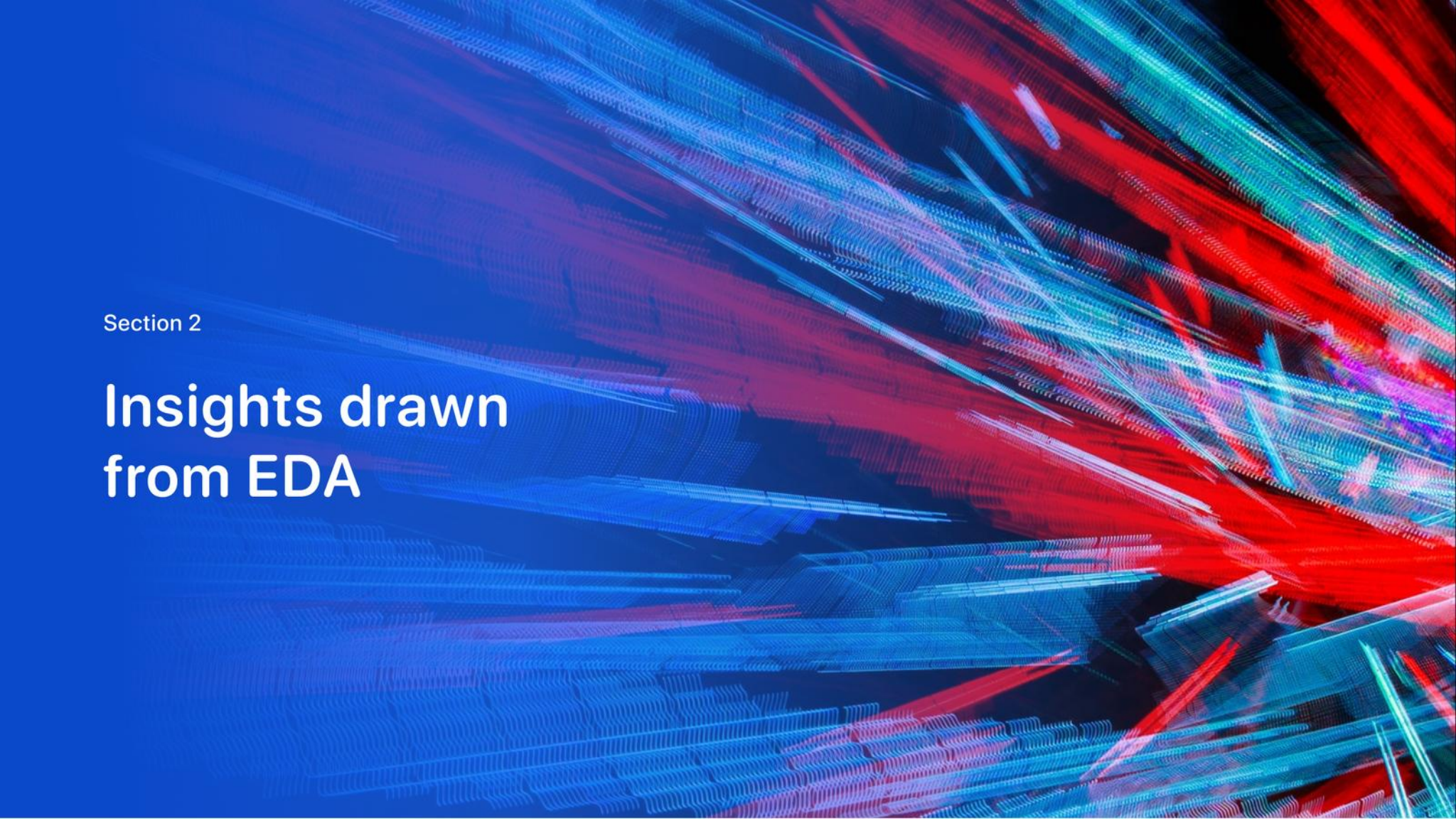
- A separate Dash notebook was not provided in this workspace.
- Equivalent dashboard-style static views were created from the completed data.
- Views answer success-count, success-ratio, and payload/outcome questions.

Predictive Analysis (Classification)

- Standardized features with StandardScaler.
- Split data 80/20 with random_state=2.
- Tuned Logistic Regression, SVM, Decision Tree, and KNN models.
- Compared CV accuracy, test accuracy, and confusion matrix behavior.

Results

- Exploratory data analysis results.
- Interactive map and dashboard-style analytics.
- Predictive analysis results.

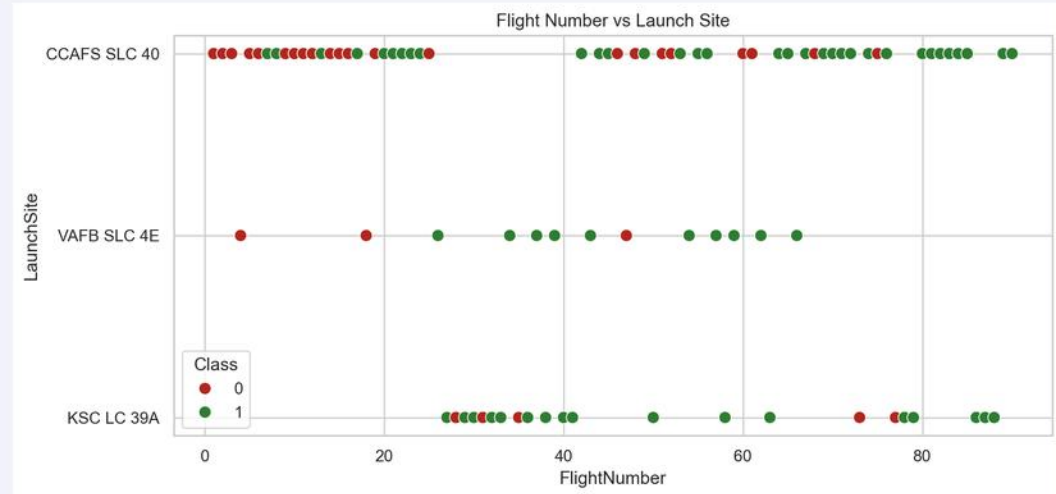
The background of the slide is an abstract composition. It features a solid blue area on the left side, which transitions into a complex pattern of diagonal streaks in shades of blue, red, and cyan on the right. These streaks have a textured, almost woven appearance, suggesting a digital or data-driven theme. The overall effect is dynamic and modern.

Section 2

Insights drawn from EDA

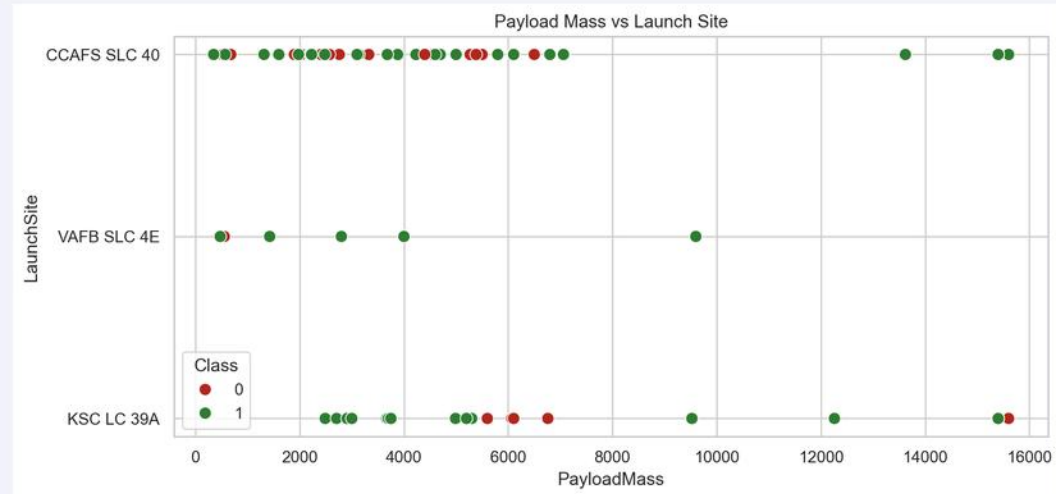
Flight Number vs. Launch Site

- Later flights show more successful landings, especially at active Florida sites.



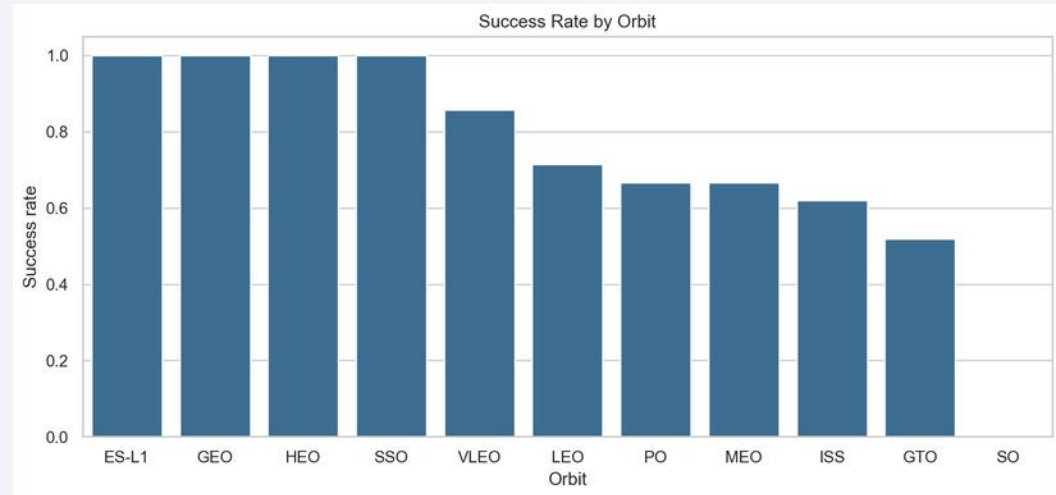
Payload vs. Launch Site

- Landing success occurs across a wide payload range; site and mission profile still matter.



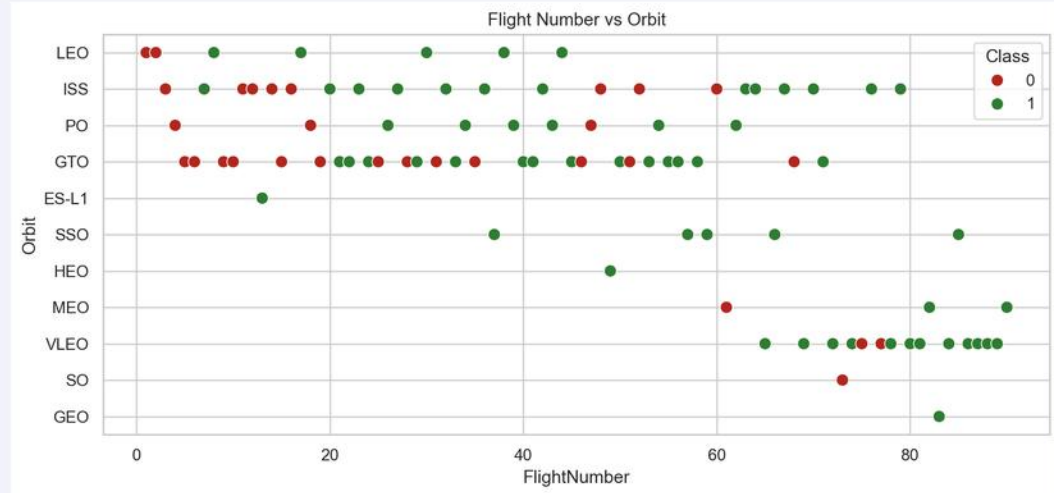
Success Rate vs. Orbit Type

- Low-frequency orbit types show high success in this sample; GTO missions are more challenging.



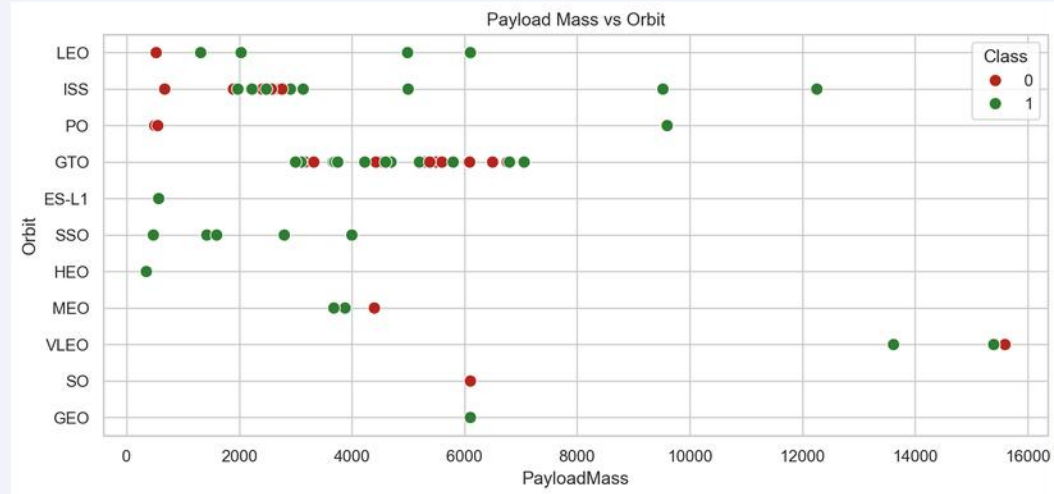
Flight Number vs. Orbit Type

- Success becomes more common in later launches across many orbit categories.



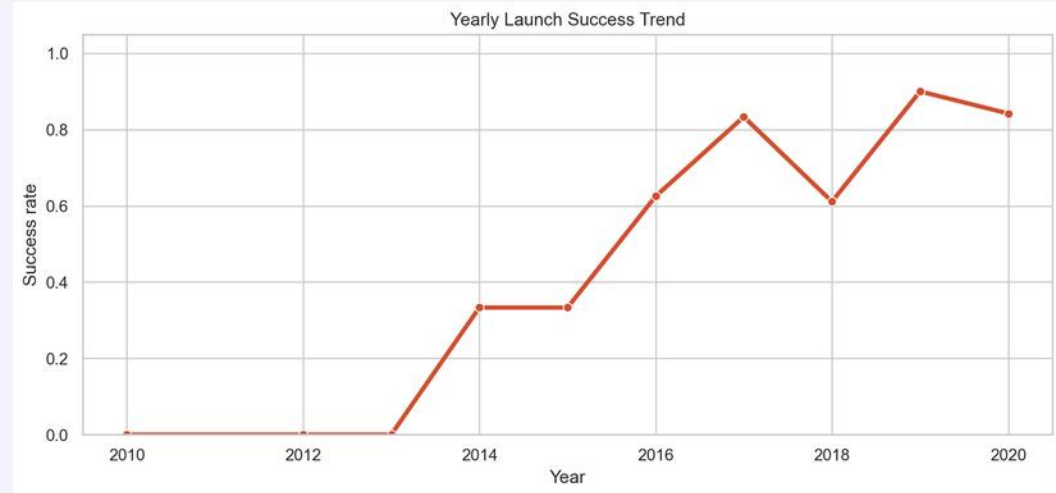
Payload vs. Orbit Type

- Payload mass varies by orbit; GTO and ISS missions span broad mass ranges.



Launch Success Yearly Trend

- Yearly success rises from early failures to consistently high rates by 2017-2020.



All Launch Site Names

Launch Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

- Four launch-site labels appear in the SQL data.

Launch Site Names Begin with 'CCA'

Date	Booster	Site	Payload
2010-06-04	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft
2010-12-08	F9 v1.0 B0004	CCAFS LC-40	Dragon demo
2012-05-22	F9 v1.0 B0005	CCAFS LC-40	Dragon C2+
2012-10-08	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1
2013-03-01	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2

- The first five CCA records are early CCAFS LC-40 launches.

Total Payload Mass

Metric	Value
NASA (CRS) total payload	45,596 kg

- NASA (CRS) launches carried 45,596 kg in total payload mass.

Average Payload Mass by F9 v1.1

Metric	Value
F9 v1.1 average payload	2,928.4 kg

- F9 v1.1 launches averaged 2,928.4 kg of payload.

First Successful Ground Landing Date

Metric	Date
First successful ground-pad la	2015-12-22

- The first successful ground-pad landing was 2015-12-22.

Successful Drone Ship Landing with Payload between 4000 and 6000

Booster Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

- Four booster-version records match the filter.

Total Number of Successful and Failure Mission Outcomes

Mission Outcome	Count
Success	98
Success (payload unclear)	1
Success	1
Failure (in flight)	1

- Mission outcomes are overwhelmingly successful.

Boosters Carried Maximum Payload

Booster Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2

- Multiple Block 5 boosters carried maximum payload mass.

2015 Launch Records

Month	Landing Outcome	Booster	Site
January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

- January and April 2015 include failed drone-ship landing outcomes.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Landing Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

- No attempt is most frequent in the early period.

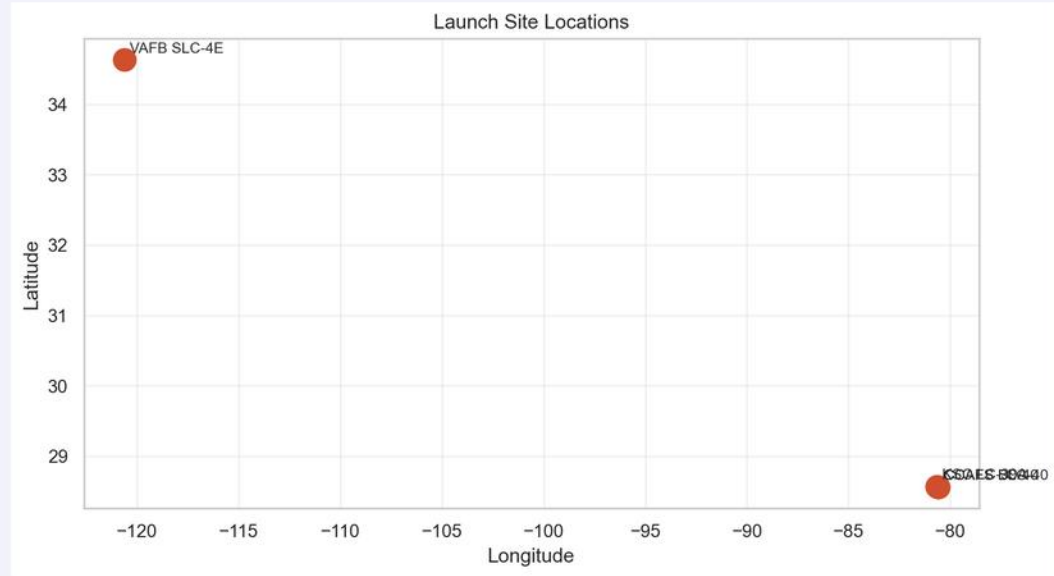
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 4

Launch Sites Proximities Analysis

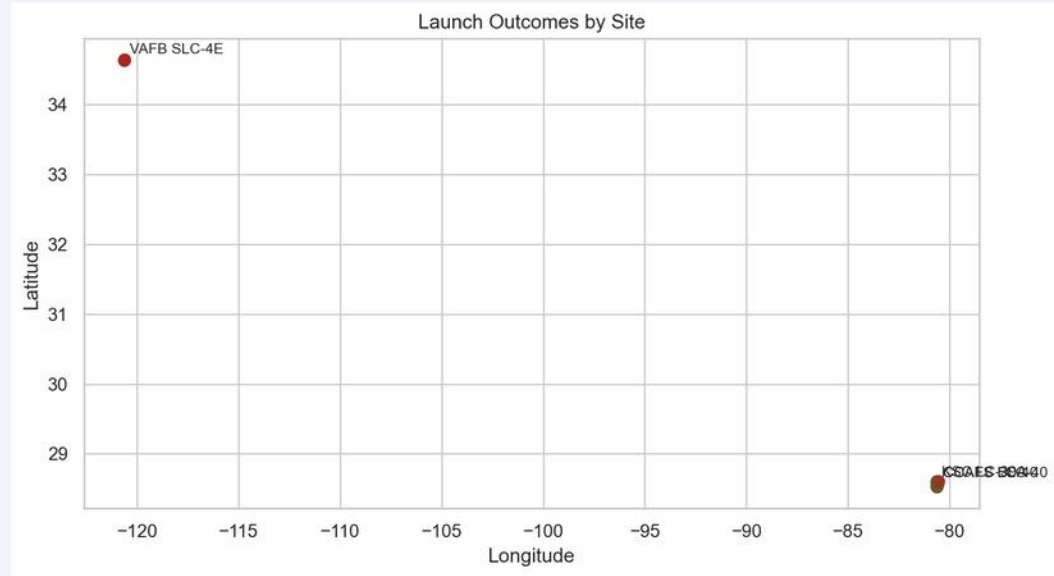
Launch Site Location Markers

- All launch sites are coastal and at relatively low latitudes.



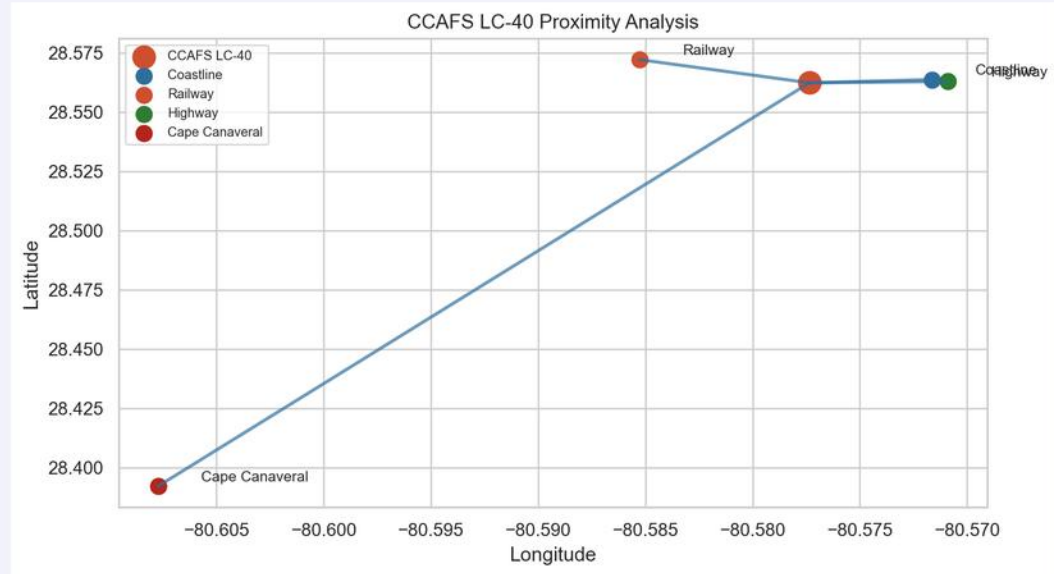
Color-Labeled Launch Outcomes

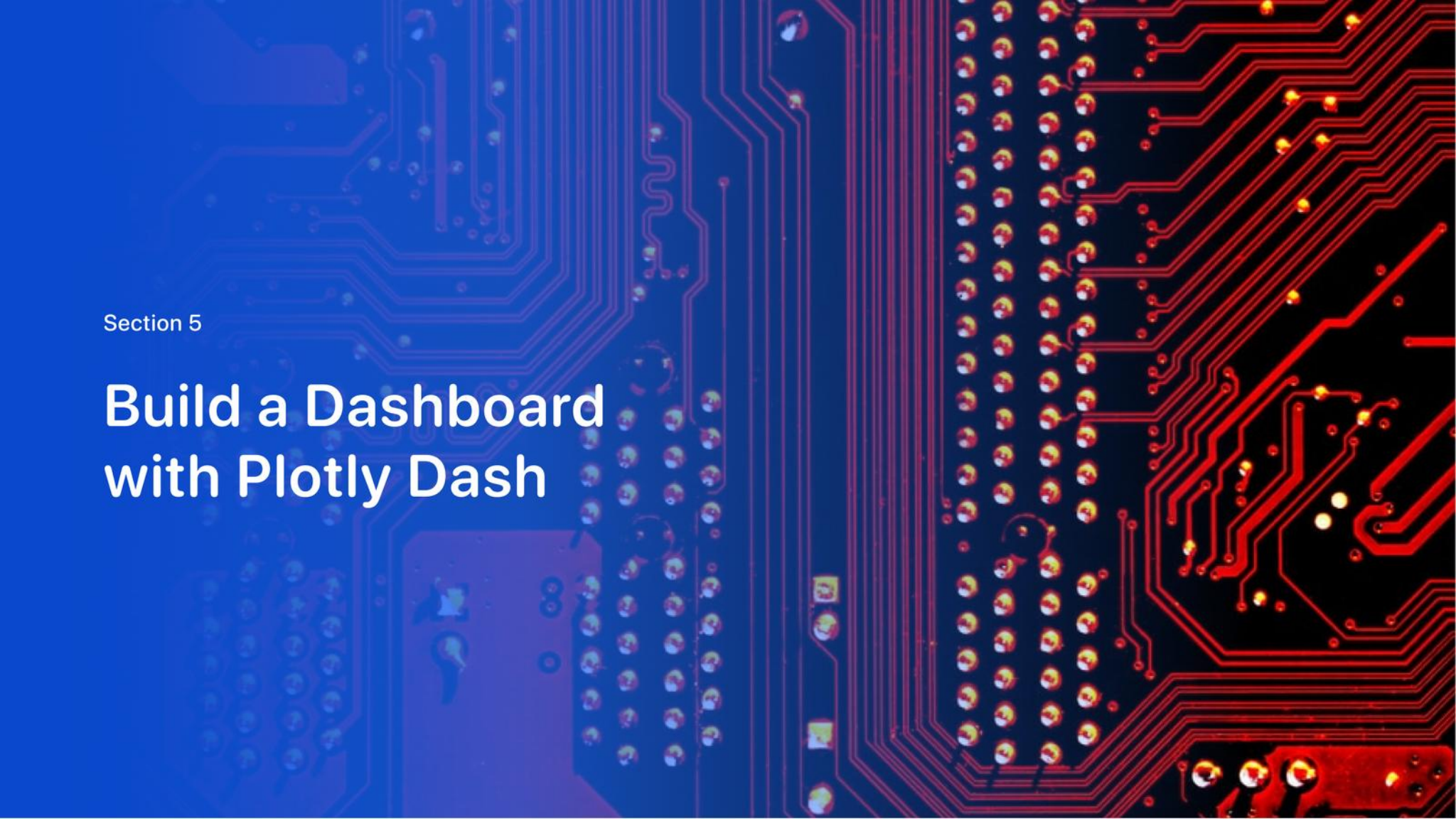
- Marker clusters show outcomes at the four launch sites.



Launch Site Proximities

- CCAFS LC-40 is close to coastline and transport routes.



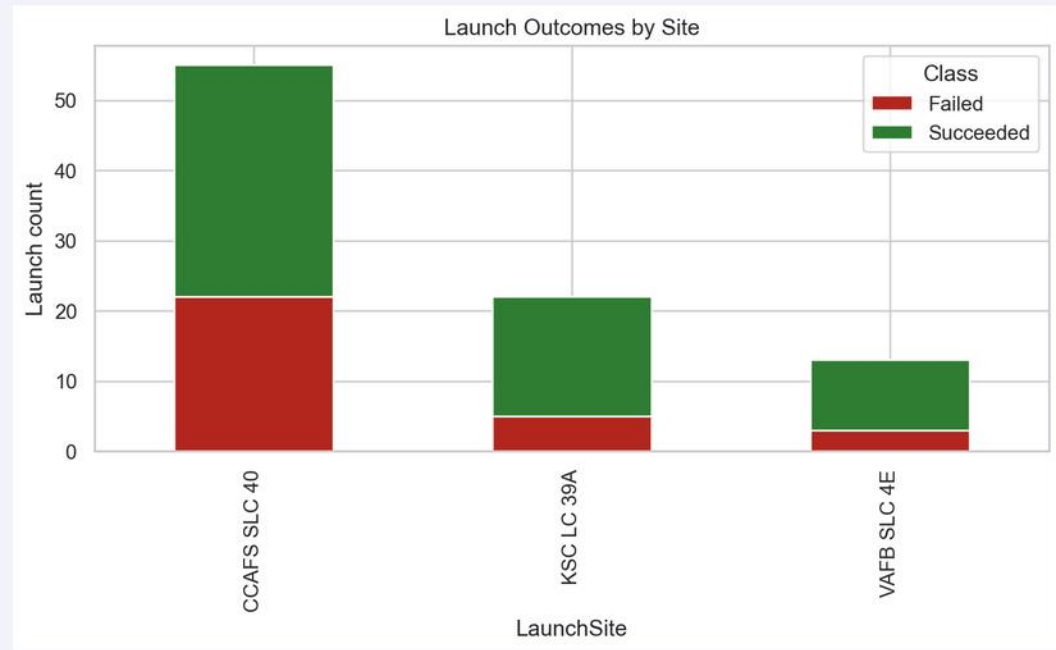


Section 5

Build a Dashboard with Plotly Dash

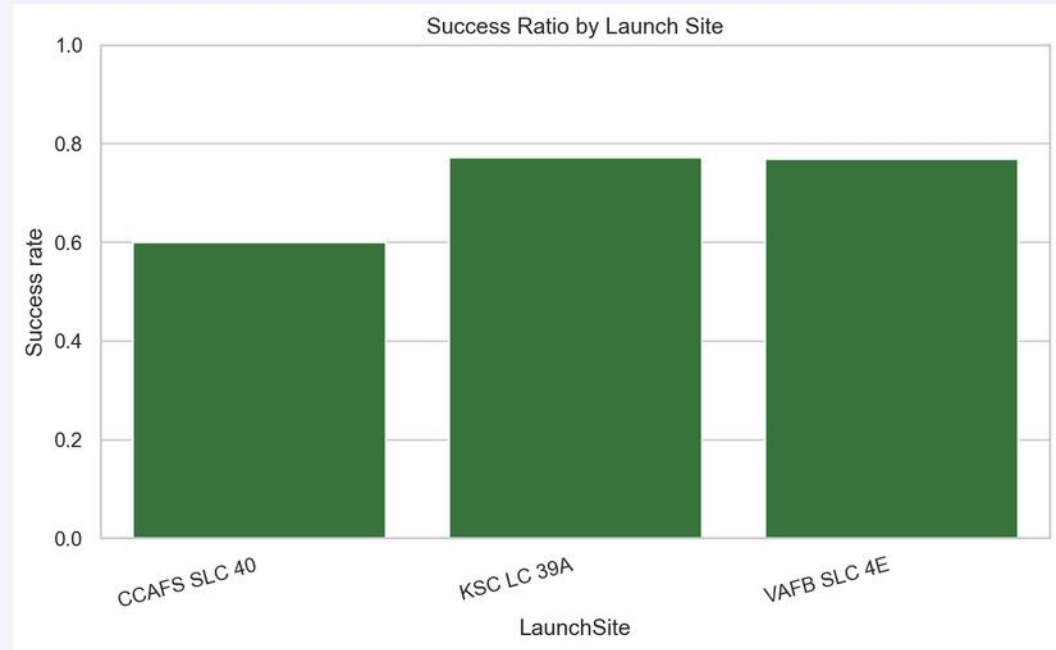
Launch Success Count by Site

- CCAFS SLC 40 has the most launches.



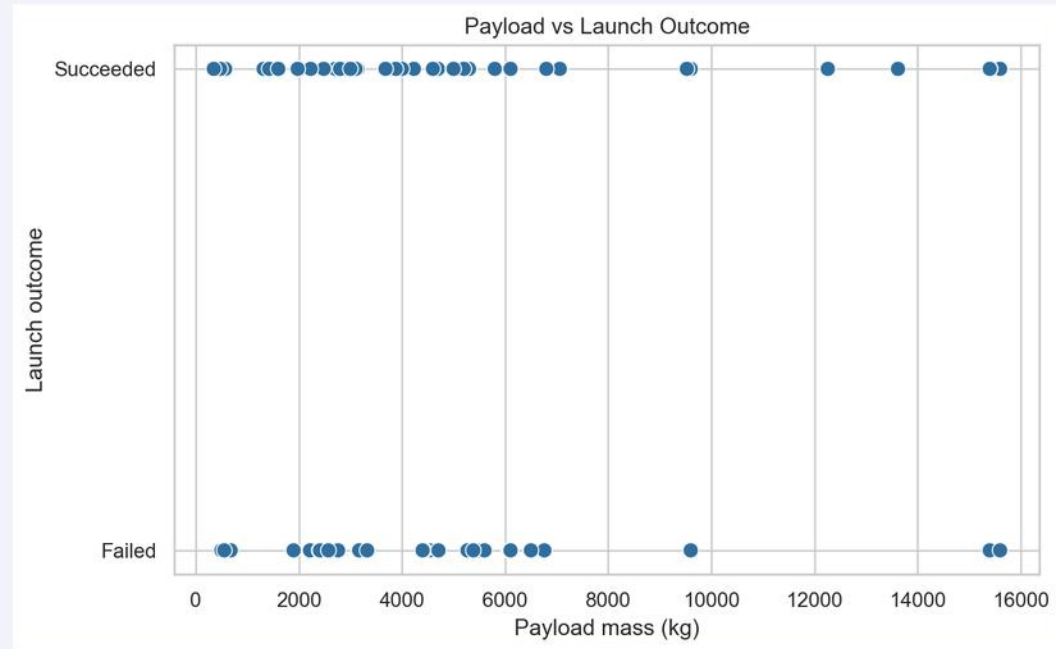
Launch Site with Highest Success Ratio

- KSC LC 39A has the highest success ratio among main sites.



Payload vs. Launch Outcome

- Payload alone is not enough to explain outcome.



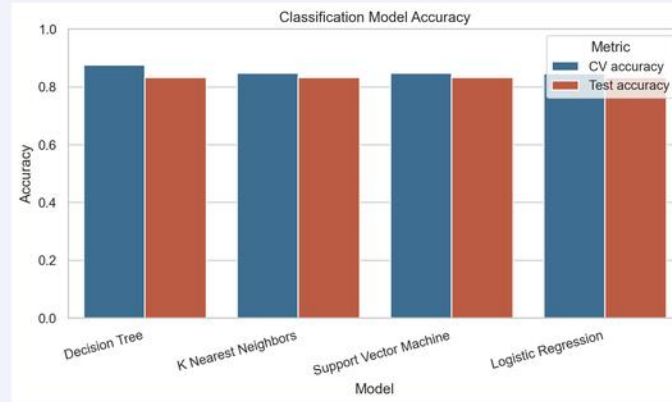


Section 6

Predictive Analysis (Classification)

Classification Accuracy

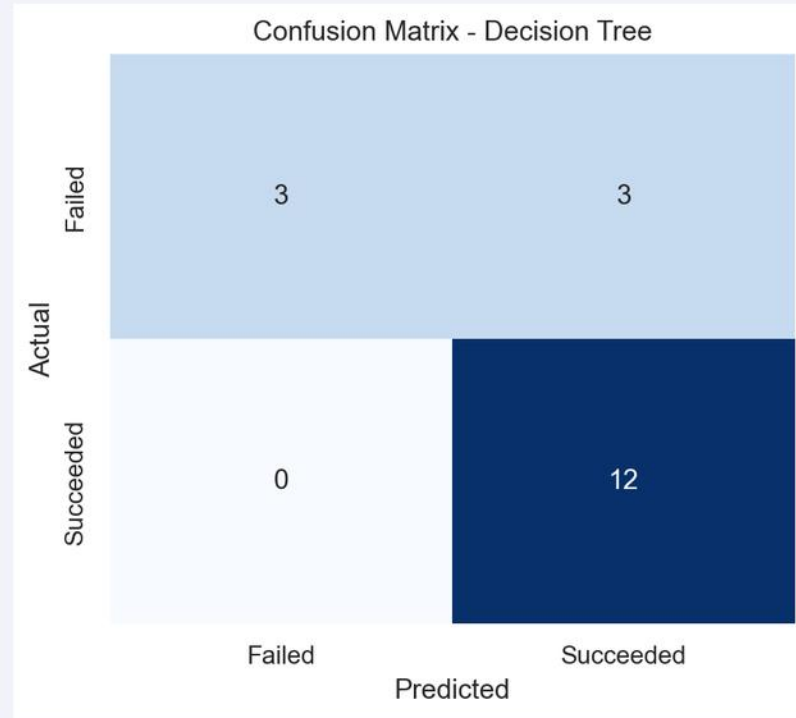
- Decision Tree had the strongest CV score; all models tied at 83.3% test accuracy.



Model	CV acc.	Test acc.
Decision Tree	0.877	0.833
K Nearest Neighbors	0.848	0.833
Support Vector Machine	0.846	0.833
Logistic Regression	0.846	0.833

Confusion Matrix

- The best model identified all 12 successful test landings.
- The remaining errors are false positives.



Conclusions

- Launch success improved materially over the observed period.
- Orbit, launch site, flight number, and booster/pad indicators are useful predictors.
- Reusable first-stage outcomes can be predicted with moderate accuracy.
- Richer recent data would likely improve production prediction.

Appendix

- Completed notebooks are saved in their original module folders.
- Generated charts and tables are saved in Module 5/report_assets.
- The root PowerPoint template is filled while preserving its original theme.
- Replace local-reference notes with a GitHub URL after upload if required.

References

- IBM Data Science Capstone course notebooks and provided SpaceX data sets.
- SpaceX public launch API static course snapshot.
- Wikipedia Falcon 9 and Falcon Heavy launch table snapshot.
- Python libraries: pandas, seaborn, matplotlib, folium, SQLite, and scikit-learn.